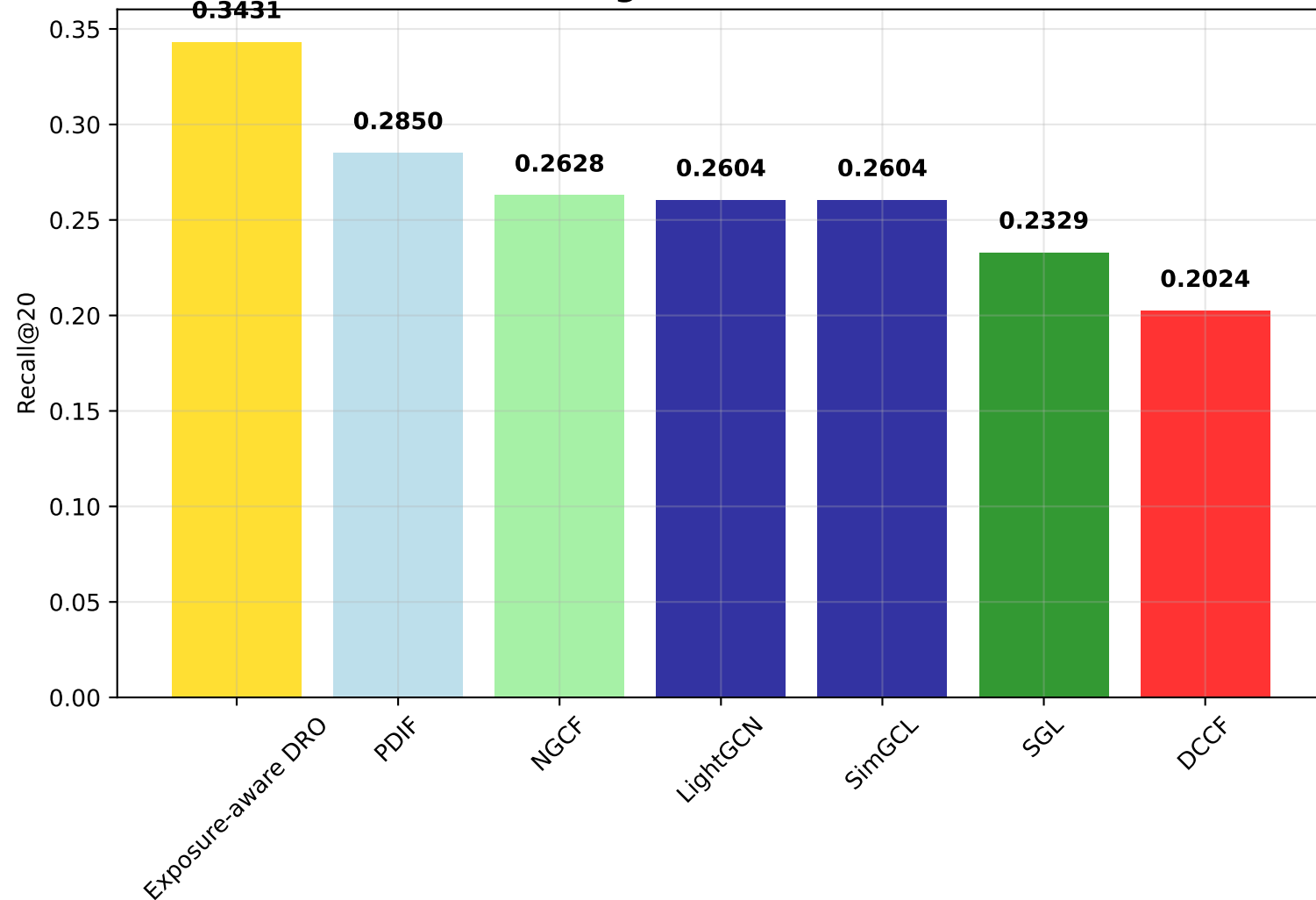
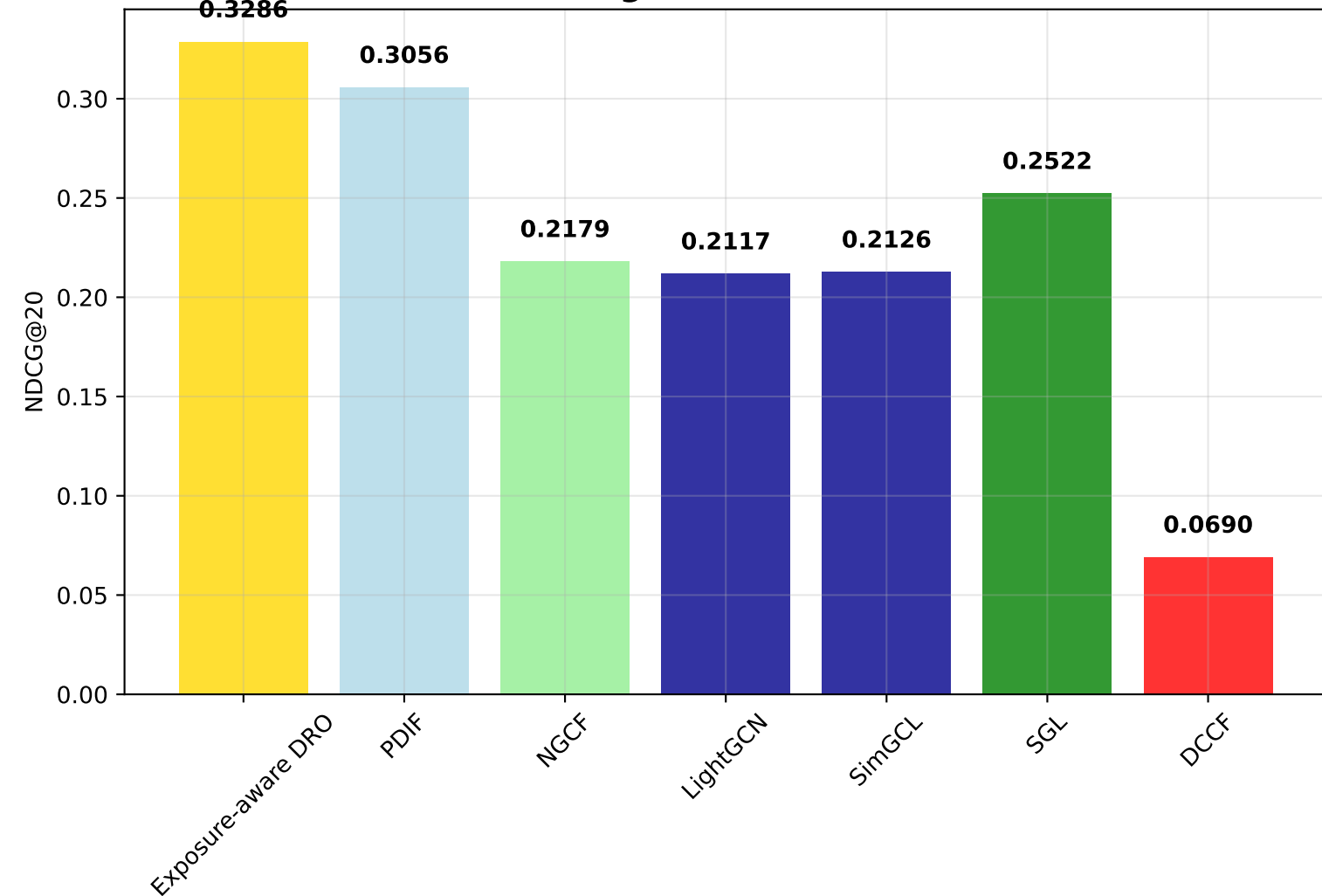


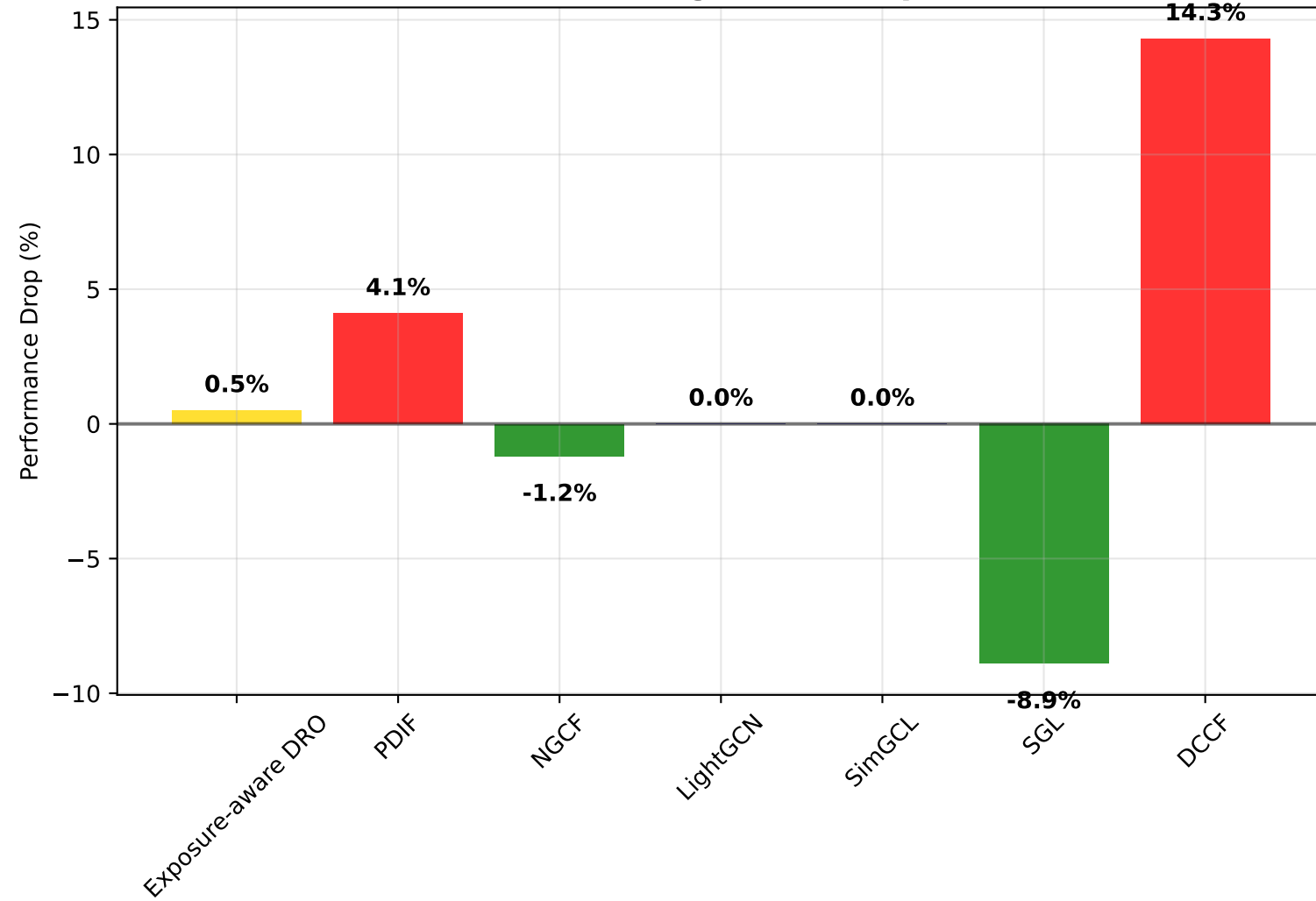
Performance Comparison: Recall@20
(Higher = Better)



Ranking Quality: NDCG@20
(Higher = Better)



Robustness Analysis: Performance Drop %
(Lower = Better, Negative = Improvement)



BREAKTHROUGH DISCOVERIES:

- OVERALL CHAMPION:
Exposure-aware DRO (0.3431 recall, 0.5% drop)
- PERFECT ROBUSTNESS:
LightGCN & SimGCL (0% degradation)
- COUNTER-INTUITIVE IMPROVEMENTS:
 - SGL: -8.9% (major improvement under noise)
 - NGCF: -1.2% (improves under noise)
- MOST VULNERABLE:
DCCF: 14.3% drop (needs solution)
- RESEARCH IMPACT:
 - 7 models evaluated (2019-2025)
 - 42 total experiments
 - First comprehensive comparative study
 - Perfect robustness phenomenon discovered